Which Grounding Prevents Stereotype Reliance? A Three-Arm Audit of LLM-Simulated Regional Humans

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Large language models are increasingly used to simulate the opinions of people from specific countries, yet bare country-label prompting is known to amplify cultural stereotypes. We audit *which grounding method* prevents this failure on a small open-weight model (Qwen3-1.7B, CPU-only): (1) bare country labels, (2) random-individual grounding — individual survey records in the style of ValuesRAG and calibrated value personas, but drawn by uniform random sampling rather than query-conditioned retrieval — and (3) third-person forecasting. Across 3 arms × 9 countries × 12 World Values Survey items × 12 generations per cell, including 144 neutral generations (4,032 generations in the full experiment), grounded in GlobalOpinionQA human distributions, we find a disciplined negative result: neither random-individual grounding nor forecasting significantly improves distributional fidelity over bare labels (Wasserstein 0.491 [0.42, 0.56] and 0.517 [0.45, 0.59] vs. 0.545 [0.47, 0.63]; $p = 0.089$ (Holm-adjusted $p = 0.179$, the preregistered decision rule) and $p = 0.301$), though the random-individual arm shows higher within-cell dispersion (entropy ratio 1.21 [1.09, 1.37] vs. 1.10 [0.98, 1.24] (medians 0.95, 1.01, 0.95 for label, random-individual, forecast), $p = 0.0018$; partly mechanical, since the individual profiles summarize background items correlated with the targets). Country separability over per-generation profiles is at chance on the label (0.102 [0.05, 0.16]) and forecast (0.111 [0.06, 0.18]) arms but modestly above it on the random-individual arm (0.213 [0.14, 0.29]; 1-NN leave-one-out accuracy vs. chance 0.111; random-individual permutation $p = 0.0032$) — the reverse of the preregistered H4, though far below the separable cultural caricatures documented on frontier models; on this small model the binding distortion is instead severe *flattening* in the single-item pilot (mean entropy ratio 0.35, range 0.0–1.18 over 16 cells), which relaxes in the multi-item format (ratios 1.10–1.21 (medians 0.95, 1.01, 0.95), 3.7% of label cells severely flattened). We release a one-command reproduction package with all prompts, code, and analysis (package documentation and deviation log in `repro/README.md`).

Additional Key Words and Phrases: LLM simulation, cultural alignment, stereotype audit, World Values Survey, evaluation

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1 INTRODUCTION

Simulating the opinions of people from specific countries with large language models (LLMs) has become a common evaluation and application paradigm, from cross-cultural alignment benchmarks to synthetic survey respondents [2, 6, 8]. The cheapest way to steer a model toward a regional population — appending a bare country label such as "You are a typical person from Japan" — is also the most suspect: Durmus et al. [6] show that country-perspective prompting shifts responses toward the target population but can reflect harmful cultural stereotypes, and a growing literature documents the resulting distortions. Cheng et al. [4, 5] formalize *caricature* as low individuation combined with high exaggeration; we use the term in both senses it takes in the literature: highly separable cultural prototypes on frontier models, and low-entropy

flattening on small models. Wdowicz [12] finds GPT-4.1-mini produces cultural profiles that are 99.8% k-NN separable with near-zero within-culture variance; Wang et al. [11] show LLM persona replacement flattens within-group heterogeneity against ∼3,200 human respondents.

Several grounding methods have been proposed to prevent this failure. Seo et al. [9] retrieve per-individual value summaries from the World Values Survey (WVS) instead of relying on model priors (ValuesRAG), explicitly arguing that role assignment invites stereotypes. Abels et al. [1] construct calibrated value personas from individual-level WVS responses, showing reduced prediction error with the largest gains in underrepresented populations. Wedgwood et al. [13] compare elicitation frames and find third-person forecasting aligns best with matched human distributions. Yet no study has compared these preventions head-to-head with caricature metrics on the small open-weight models that independent auditors can actually run — the setting where stereotype reliance may not even be the binding failure mode.

We fill this gap with a preregistered three-arm audit on Qwen3-1.7B (CPU inference only). Our contributions: (1) a head-to-head comparison of bare country labels, random-individual grounding, and third-person forecasting, using identical items, countries, and sampling budgets; (2) an evaluation battery combining distributional fidelity (Wasserstein distance to GlobalOpinionQA human distributions), a flattening check (entropy ratio, following Wang et al.), and caricature instruments adapted from CoMPosT and Wdowicz (k-NN separability, individuation, exaggeration); (3) evidence that on a small model separable caricature does not materialize as on frontier models (country separability at chance on two arms, modestly above chance on the random-individual arm — the reverse of the preregistered direction) while the binding distortion depends on format — severe flattening in single-item prompts, country-invariant but variable answers in multi-item prompts — with implications for how audits should be designed across model scales; (4) a one-command reproduction package.

2 RELATED WORK

Simulating human opinions with LLMs. Argyle et al. [2] introduce silicon sampling, conditioning models on demographic backstories to recover subgroup response distributions. Santurkar et al. [8] (OpinionQA) show model defaults reflect a narrow demographic and that conditioning moves answers unevenly. Tao et al. [10] map model responses onto the Inglehart–Welzel cultural map against WVS data, finding national role-assignment reduces but does not eliminate bias. Bulté and Rigouts Terryn [3] test 10 LLMs on 63 cultural items and find explicit cultural framing beats prompt language, but a systematic bias toward a restricted country set persists. Mohammadi and Bagheri [7] document persistent WEIRD bias across 26 models. Zhou et al. [15] caution that naive de-WEIRDing is not automatically safer. We take no position on whether LLMs *should* simulate humans; we audit *how* the grounding choice changes stereotype reliance when they do.

Documenting stereotype reliance and caricature. Durmus et al. [6] build GlobalOpinionQA from WVS and Pew surveys and show country prompting can amplify stereotypes. Cheng et al. [4] (Marked Personas) find model portrayals contain higher rates of stereotypes than human-written ones; Cheng et al. [5] (CoMPosT) define caricature via individuation and exaggeration and find GPT-4 simulations highly susceptible. Wdowicz [12] reports near-perfect separability of prompted cultural profiles and a 75% anglocentric default

in neutral conditions. Wang et al. [11] demonstrate flattening of within-group heterogeneity. Our audit adapts these instruments to a survey-response setting.

Grounding methods that may prevent stereotype reliance. Seo et al. [9] (ValuesRAG) ground responses in retrieved individual WVS value summaries, outperforming role-assignment baselines. Abels et al. [1] sample value personas from target populations and add a mean-preserving calibration for LLM under-dispersion. Wedgwood et al. [13] find third-person forecasting yields the strongest directional alignment across elicitation frames. Our three arms approximate these three proposals under one protocol — with the arm-2 caveat that we sample individuals uniformly at random rather than performing query-conditioned retrieval as in ValuesRAG — which no prior work has done.

Related recent work. Zhong et al. [14] (EMNLP 2026 Findings) study demographic-profile conditioning on WVS items across seven models, finding that profiles improve value alignment at a systematic cost to individuality ("alignment by stereotyping"). Their work examines the alignment–individuality tradeoff of profile conditioning; ours instead compares three grounding *methods* head-to-head with caricature instruments on a small open model, asking which grounding best prevents stereotype reliance. Where their findings overlap with ours — e.g., whether individual grounding preserves within-population variation — we discuss the agreement explicitly in the Discussion.

3 METHOD

3.1 Arms

All arms use Qwen3-1.7B (Q4_K_M GGUF — a 4-bit quantized model file, llama.cpp, CPU-only, 2 threads). Each generation is grammar-constrained to emit exactly the option letters; sampling uses temperature 0.7, top-p 0.9, and fixed seeds. In the full experiment, each prompt presents all 12 survey items (randomized order) and the model returns 12 comma-separated letters; prompts share a common structure and only the grounding frame differs:

- **Arm 1 – bare country label:** *"You are a typical person from {country}. Answer each question below the way they would."*
- **Arm 2 – random-individual grounding:** *"Below is the profile of a real survey respondent from {country}: {profile}. Answer each question below from this person's perspective."* Each prompt conditions on a fresh respondent drawn uniformly from a pool of individual response summaries (natural-language renderings provided by the redistribution; see Section 4); the profile summarizes background items only, never the target questions. This is in the style of ValuesRAG but with uniform random draws rather than query-conditioned retrieval.
- **Arm 3 – third-person forecasting:** *"Predict how a typical person from {country} would answer each of the questions below."*
- **Arm 0 – neutral:** no country cue (12 seeds).

3.2 Metrics

Let $H(c, i)$ be the human response distribution for country c and item i, and $M(a, c, i)$ the model's empirical distribution over 12 generations.

- **Distributional fidelity:** 1-Wasserstein distance $W(M(a,c,i), H(c,i))$ on the option ordinal scale, averaged over cells.
- **Flattening:** entropy ratio $R = \mathrm{H}(M)/\mathrm{H}(H)$ per cell; $R < 1$ indicates flattening [11]. We report mean R and the fraction of cells with $R < 0.5$ (severe flattening).
- **k-NN separability:** one profile per generation (108 per arm: 9 countries × 12 generations), each the 12-element vector of option indices; we report 1-NN leave-one-out accuracy at predicting the country (Hamming distance), chance $1/9 \approx 0.111$ [12]. The human baseline applies the same computation to 12 synthetic respondents per country, each drawn independently from that country's human marginals. This uses a seeded RNG, not real human data: it measures whether the human distributions themselves are country-separable at this granularity. (A modal-profile variant with one profile per country is degenerate — leave-one-out can never predict the correct country — and was dropped; see Results.)
- **Individuation:** within-country variance of per-generation profiles divided by total variance (adapted from [5]); low values indicate caricature.
- **Exaggeration:** mean distance between model and human country profiles divided by mean pairwise distance between human country profiles (adapted from [5]); values above 1 indicate the model pushes countries further out than real countries differ.

3.3 Hypotheses (preregistered)

H1 (replication): arm-1 k-NN separability $\geq 90\%$, mean $R < 0.7$, and neutral generations match the anglosphere modal profile $\geq 60\%$ of the time. H2 (primary): arms 2 and 3 have lower mean Wasserstein distance than arm 1 (paired tests over cells, Holm correction). H3: arms 2 and 3 show higher mean entropy ratio than arm 1 (less flattening). H4: k-NN separability is lower for arms 2 and 3 than arm 1.

4 EXPERIMENTAL SETUP

Items and countries. 12 WVS items from GlobalOpinionQA [6] (environment vs. growth, immigration policy, freedom vs. security, willingness to fight for country, human-rights respect, political interest, job priority for natives, social trust, women's electoral opportunities, local voting, science vs. religion, voter bribery), each with 2–4 substantive options, each covering the same 9 countries: Brazil, Egypt, Germany, Great Britain, Indonesia, Japan, Mexico, Nigeria, Turkey. "Don't know"/refusal options are excluded from prompts; human distributions are renormalized over substantive options. India was dropped during item selection for lack of GlobalOpinionQA coverage on the selected items, and Great Britain replaced the United States (see preregistration Section 10 for the deviation log).

Individual-level pool (arm 2). We use the `oxford-llms/world_values_survey_2017_2022_sft` public redistribution (formatted for supervised fine-tuning, SFT) of WVS Wave 7 individual responses (405,729 respondent-question records after ethics screening; see Ethics section). WVS's own trend file requires registration that blocks machine access; we disclose the full provenance chain. Each record is a natural-language summary of one respondent's background answers; the target question is never part of the profile (no leakage).

Protocol. Preregistration (arms, items, metrics, tests, decision rules, ethics constraint) was frozen before any model run; deviations are logged in the reproduction package. We first ran a single-item replication pilot (4 countries × 4 items, arms 0–1, 20 generations each) to test H1, then the full experiment: 336 prompts, each presenting all 12 items (order randomized per prompt, recorded) with the model returning 12 comma-separated option letters under a position-specific grammar. Arm 1: 9 countries × 12 seeds; arm 2: 9 countries × 12 fresh individual profiles; arm 3: 9 countries × 12 seeds; arm 0: 12 seeds with no country cue. This yields 12 answers per (arm, country, item) cell (4,032 records). Because all 12 items share a single prompt, answers may be shaped by cross-item context effects, which the per-prompt order randomization only partially mitigates.

5 RESULTS

5.1 Replication pilot (H1)

The single-item pilot (4 countries × 4 items, arms 0–1, 20 generations per (country, item) cell; arm 0 ran 80 generations per item with no country cue, matching arm 1's 4×20 per-item budget) does not replicate the caricature pattern documented on frontier models. The preregistered separability computation was degenerate under the pilot design and is not reported. The informative pilot result is the mean entropy ratio of 0.35, confirming severe flattening (H1 required < 0.70; 9 of 16 pilot cells severely flattened), with near-deterministic, country-invariant answers (e.g., all 20 generations identical on the immigration item). Neutral generations match the Great Britain modal profile only 35.0% of the time (computed on the 320 pilot neutral generations; H1 required $\geq 60\%$; per-item matches range 0.0–1.0, so the pooled figure masks item-level heterogeneity). Per the preregistration, the three-arm comparison proceeds but is framed as exploratory.

5.2 Three-arm comparison

Across 108 (country, item) cells, mean Wasserstein distances to human distributions are 0.545 [0.47, 0.63] (label), 0.491 [0.42, 0.56] (random individual), and 0.517 [0.45, 0.59] (forecast). Neither grounding arm significantly improves on bare labels: H2a (random-individual < label) $p = 0.089$ (Holm-adjusted $p = 0.179$, $d = 0.17$); H2b (forecast < label) $p = 0.301$ ($d = 0.10$). **H2 is not supported.**

Mean entropy ratios are 1.10 [0.98, 1.24] (label), 1.21 [1.09, 1.37] (random individual), and 1.12 [1.02, 1.23] (forecast) (medians 0.95, 1.01, 0.95). The random-individual arm shows a higher mean entropy ratio than bare labels (H3a: Wilcoxon $p = 0.0018$, rank-based $r = 0.30$, standardized mean of paired differences $d = 0.31$), i.e. higher within-cell dispersion; forecasting does not (H3b: $p = 0.438$, $d = 0.04$). The H3a difference survives a country-level cluster bootstrap (mean paired difference 0.11, 95% CI [0.05, 0.18], two-sided $p < 0.001$). This is a descriptive dispersion difference, not evidence of debiasing: the arm-2 profiles summarize background items that correlate with the target questions, so part of the dispersion gain is mechanical. On the $|\log R|$ distance from a perfectly calibrated ratio of 1, the random-individual arm is if anything closer to 1 (mean 0.25 vs. 0.31 for the label arm; paired difference -0.06, 95% CI $[-0.13, 0.00]$, $p = 0.059$) — the observed dispersion gain is bounded: it is not explained by overshoot alone. Severe flattening ($R < 0.5$) affects 3.7% [0.9, 7.4] of label cells; the cross-arm differences are not significant (Fisher

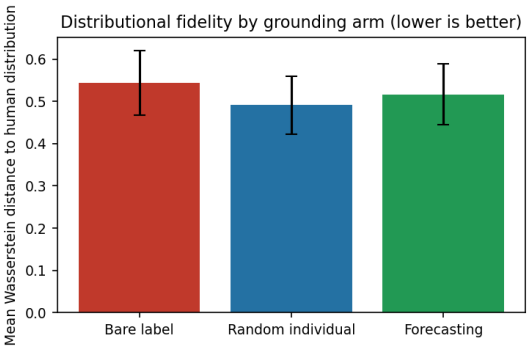


Fig. 1. Per-cell Wasserstein distance to human distributions by arm. The random-individual arm trends lower than bare labels, but the difference is not significant (H2a $p = 0.089$).

exact $p = 0.37$ vs. random individual, $p = 0.68$ vs. forecast). **H3 is partially supported** (random-individual grounding only), with the confound above. Holm correction was preregistered only for H2; H3a's $p = 0.0018$ survives any 2-test multiplicity control regardless. A sensitivity check on $|\log R|$ (distance from 1, one cell winsorized at 1.96) shows no significant overshoot (diff -0.06, 95% CI $[-0.13, 0.00]$, $p = 0.059$), bounding the mechanical component of the dispersion gain. The 108 cells cluster within countries (12 cells per country share the same human-distribution structure); tests are paired over cells, and the clustering is common to all arms, so the arm comparisons remain fair.

k-NN country separability over 108 per-generation profiles per arm is 0.102 [0.05, 0.16] (label), 0.213 [0.14, 0.29] (random individual), and 0.111 [0.06, 0.18] (forecast), against chance $1/9 \approx 0.111$ and 0.31 (range 0.23–0.36 over 20 seeds) for synthetic respondents drawn from the human marginals — a marginal-only ceiling that ignores covariance across items (Figure 3). Because the 12 generations per country share prompt and country structure, the bootstrap CIs are not cluster-robust; the permutation tests serve as the robustness check. Random-individual separability significantly exceeds both chance (permutation $p = 0.0032$; binomial 23/108 vs. 1/9, $p = 0.0016$) and the bare-label arm (arm-label permutation on the pooled profiles, $p = 0.025$) — the reverse of preregistered H4, which predicted lower separability under grounding. **H4 is reversed**, not merely null: country identity is most recoverable from the random-individual arm's outputs, consistent with individual profiles carrying country-distinctive background content. The per-item fidelity breakdown (Figure 4) is descriptive, with no multiplicity control, and suggests the random-individual arm's small edge is concentrated on specific items (e.g., item06, political interest) rather than spread uniformly.

Individuation is high across arms (0.926 [0.91, 0.96]–0.944 [0.94, 0.97]), and exaggeration ranges from 1.24 [1.11, 1.91] (random individual) to 1.47 [1.42, 2.15] (forecast). Because the underlying per-generation profiles are near chance-separable, the exaggeration reading is suggestive rather than definitive: the model pushes country modal profiles modestly further apart than real countries differ, without making individual generations country-separable.

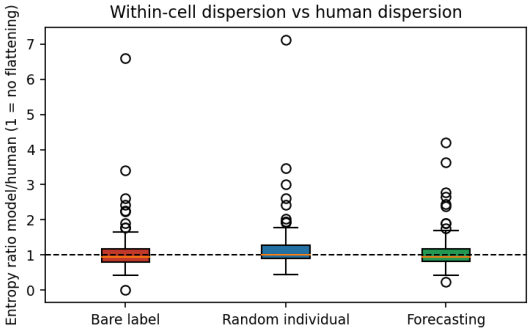


Fig. 2. Per-cell entropy ratio (model/human) by arm. The random-individual arm shows a higher ratio than bare labels (H3a $p = 0.0018$), i.e., higher within-cell dispersion (partly mechanical; see Results).

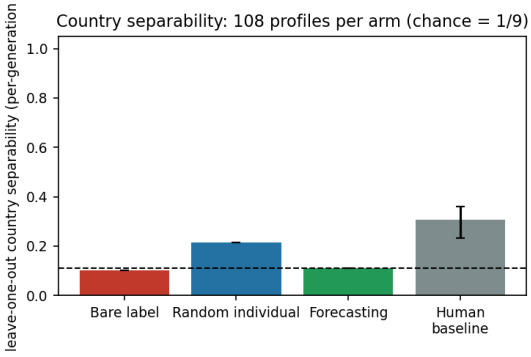


Fig. 3. 1-NN leave-one-out country separability over 108 per-generation profiles per arm (chance $1/9 \approx 0.111$; dashed line). The label and forecast arms sit at chance; the random-individual arm is modestly above it. Synthetic respondents drawn from the human marginals — a marginal-only ceiling that ignores covariance — reach 0.31 (mean over 20 seeds; range 0.23–0.36).

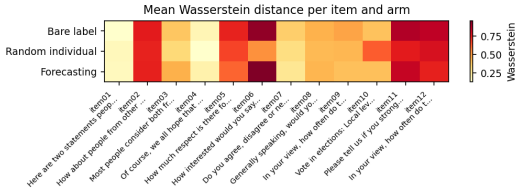


Fig. 4. Mean Wasserstein distance per item and arm, averaged over countries. The random-individual arm's fidelity edge is concentrated on specific items (e.g., item06, political interest).

6 DISCUSSION

The headline is a disciplined negative result. On a small open model, neither random-individual grounding nor third-person forecasting significantly improves distributional fidelity over bare country labels, though the random-individual arm shows higher within-cell dispersion (partly mechanical, since the profiles summarize

background items correlated with the targets). The failure mode that grounding is meant to prevent — separable cultural caricature — barely materializes: country separability over per-generation profiles sits at chance on the label (0.102) and forecast (0.111) arms and only modestly above it on the random-individual arm (0.213), the reverse of the preregistered H4. To disambiguate the reversal: above-chance separability on the random-individual arm is the expected behavior of any profile-based arm — individual profiles carry country-distinctive background content by construction — and does not by itself show that individual-based grounding harms stereotype reliance. Two regimes emerge: the single-item pilot shows severe flattening (mean entropy ratio 0.35, near-deterministic answers), while the multi-item format relaxes it (ratios 1.10–1.21, medians near 1, 3.7% of label cells severely flattened) — country-invariant but variable. This inverts the usual audit logic: on Qwen3-1.7B the risk is not that the model leans on country stereotypes too hard, but that it barely conditions on the country cue at all.

Three implications follow. First, caricature severity appears scale-dependent: with per-generation profiles, country separability on Qwen3-1.7B sits at chance on the label and forecast arms and only modestly above it on the random-individual arm (0.102–0.213 across arms vs. chance $1/9 \approx 0.11$), whereas Wdowicz [12] reports 99.8% on GPT-4.1-mini using modal profiles — the two metrics are not directly comparable, and our synthetic-human baseline of 0.31 shows the human distributions themselves are far from fully separable at this granularity, so scale-dependence is not the only explanation for the frontier gap. Still, the directional contrast suggests audits must be calibrated to model scale rather than importing frontier-model findings wholesale. Second, the random-individual arm's dispersion gain (H3a) without a fidelity gain (H2a) echoes Zhong et al.'s [14] alignment–individuality tradeoff: grounding in real individuals restores variation but does not automatically move the distribution toward the human target. Third, the multi-item format itself reduced severe flattening relative to our single-item pilot (3.7% vs. 56% of pilot cells), a methodological note for future survey-simulation audits.

7 LIMITATIONS

Single 1.7B model; 9 countries; 12 items; English prompts only (Bulté and Rigouts Terryn [3] show prompt language matters); one temperature; WVS Wave 7 vintage; the multi-item format introduces cross-item context effects (order randomized to mitigate). The arm-2 uniform draw weakens the head-to-head with ValuesRAG: our intervention grounds each prompt in a randomly sampled individual rather than query-conditioned retrieval, so the honest H2 null may partly reflect intervention weakness rather than retrieval's inefficacy.

8 ETHICS STATEMENT

This study was designed under an absolute constraint: no race or ethnicity categories appear anywhere in the design, prompts, code, analysis, figures, or manuscript. Country labels are the *audited treatment* — the thing whose misuse is measured — and measured survey data is the ground truth; the design satisfies the anti-stereotyping constraint by construction. To enforce it, an automated linter scans every prompt, item, corpus record, model output, and analysis artifact for race/ethnicity tokens and fails the run on any hit; 20,802 of 426,531 corpus rows (4.9%) were excluded because their persona summaries referenced race or ethnicity (including mentions of observed racist behavior in WVS neighborhood items). No human subjects were involved; the survey records used are anonymized public data. The audit's purpose is preventive: identifying

which grounding method keeps simulated regional humans closest to measured human distributions instead of collapsing into stereotypes. Nothing in this work endorses using LLMs as replacements for real survey respondents in high-stakes decisions.

9 CONCLUSION

We audited which grounding method prevents stereotype reliance when a small open model simulates regional humans. The answer is sobering: neither random-individual grounding nor third-person forecasting significantly beats bare country labels on distributional fidelity, though the random-individual arm shows higher within-cell dispersion (partly mechanical, per the confound noted in Section 5). More fundamentally, the stereotype-reliance failure mode — separable caricature — barely appears at this scale: country separability sits at chance on the label and forecast arms and only modestly above it on the random-individual arm, the reverse of the preregistered direction. What does bind depends on format: severe flattening in the single-item pilot, country-invariant but variable answers in the multi-item format. For practitioners, the implication is to audit the failure mode that actually binds at the deployed scale and format, not the one most documented on frontier models. We release all prompts, code, and analysis in a one-command reproduction package; the package will be released via an anonymized supplementary artifact per FAccT policy (no URLs or author names in this version).

GENERATIVE AI USAGE STATEMENT

The author used the AI tool Muse (Muse Spark 1.3, Meta) for drafting, grammatical editing, and code validation. The contents are reviewed, edited, and attested for accuracy by the author.

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